Scratch-Calibrated Strokes Gained

A course-specific baseline for measuring an amateur's game against a scratch golfer

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Why a different baseline

Strokes gained measures how many strokes a shot saved relative to a baseline golfer: a shot's value is the expected strokes to hole out from where it started, minus the expected strokes from where it finished, minus one. The framework is Mark Broadie's and it is the right way to decompose a score. The question is who the baseline golfer is and what course they are playing.

Consumer apps score an amateur against a PGA Tour player, using the same tables on every course. For a player trying to reach scratch, that creates two problems.

1. **The wrong target.** The gap between scratch and tour is three to four strokes a round, concentrated in approach play. Measuring against tour tells a 9-handicap how far they are from a PGA Tour card, not from scratch.
2. **No knowledge of the course.** A 356-yard par 4 with out of bounds on both sides of a 45-yard corridor is not the same hole as a 356-yard par 4 with open rough. A scratch golfer lays up on the first; a course-blind table expects a driver and scores a correct layup as a poor shot.

This note builds the baseline for a **scratch golfer on a specific course**. The expected-strokes tables come from published scratch statistics (Shot Scope, Arccos). Tee shots on par 4s and 5s come from a decision model that knows the corridor width and where the trouble is. Hole expectations are then calibrated so the eighteen holes sum to the USGA course rating. Worked example: Fox Creek Golf Club, Smyrna GA, Green tees, 3,802 yards, par 62, rating 59.9.

Notation: $E(\text{lie}, d)$ is the expected number of strokes for the baseline golfer to hole out. Distances are in yards except on the green, where d is in feet.

1. Strokes gained

$$\text{SG} = E(\text{start}) - E(\text{end}) - 1 - \text{penalty}$$

2. Putting

From the scratch one-putt rate p_1, three-putt rate p_3 and four-putt rate p_4 at each anchor distance,

$$E_G(f) = 2 - p_1 + p_3 + p_4$$

The exact value is $2 - p_1 + p_3 + 2p_4$; the simplification is conservative by p_4, under 0.04 at any distance. Anchors are interpolated linearly; distances are rounded to the nearest foot.

Feet	p_1	p_3	E_G
3	0.94	0.000	1.06
5	0.73	0.002	1.27
10	0.36	0.012	1.65
15	0.22	0.030	1.81
20	0.14	0.055	1.92
30	0.065	0.13	2.07
40	0.035	0.20	2.17
60	0.018	0.34	2.34

Published inputs: Shot Scope 0-handicap make rates by band (two tables, which disagree at 12–18 ft; the midpoint is used); Arccos three-putt rates for 1–5 index players (9.4% from 20–25 ft, 19.6% from 25+ ft); an Arccos scratch figure of about 25% from 36–40 ft; Shot Scope scratch figures of 3% three-putts per hole and 4.6 ft left after a first putt from 30 ft.

3. Approach shots, 50–250 yards

Condition on hitting the green. With g the GIR probability, x the mean on-green proximity and E_{miss} the short-game value of a typical miss,

$$E_F(d) = 1 + g(d)\, E_G\big(\kappa\, x(d)\big) + \big(1 - g(d)\big)\, E_{\text{miss}}(d)$$

The factor $\kappa = 0.7$ corrects for the spread of proximity around its mean. E_G is concave, so by Jensen's inequality $\mathbb{E}[E_G(X)] < E_G(\mathbb{E}[X])$; using $0.7\,\bar{x}$ reproduces published tour fairway values from tour GIR and proximity to within about 0.05.

Shot Scope publishes proximity *including* misses, $\bar{x}$, by distance band. With m the mean miss distance,

$$\bar{x} = g\, x + (1 - g)\, m$$

The anchors for g and x were chosen so this holds with plausible miss distances (53, 59, 70 and 111 ft for the four bands) and an overall GIR near the published 52–59%. At 125 yards: $0.62 \times 25 + 0.38 \times 59 = 38$ ft, matching the published 38 ft for 100–150 yards, and

$$E_F(125) = 1 + 0.62\, E_G(17.5) + 0.38\, (2.55) = 1 + 0.62\, (1.87) + 0.97 \approx 3.13$$

Yards	GIR	On-green proximity (ft)	E_{miss}	E_F
50	0.80	14	2.45	2.80
100	0.68	21	2.50	3.03
150	0.55	30	2.60	3.23

Yards	GIR	On-green proximity (ft)	E_{miss}	E_F
200	0.38	42	2.72	3.47
250	0.22	55	2.90	3.64

Rough and fairway-bunker lies use the same formula with scaled inputs: $g \times 0.8$, $x \times 1.25$, $E_{\text{miss}} + 0.10$ for rough; 0.65, 1.30, +0.20 for sand. From 150 to 250 yards the value is the better of going for the green and laying up, where

$$E_{\text{layup}}(d) = 1 + 0.7\,E_F(L) + 0.3\,E_R(L), \qquad L = \max(d - 220,\ 30)$$

Beyond 250 yards only the layup branch applies.

4. Around the green and recovery

Inside 50 yards, an up-and-down is two strokes and a failure is usually three. With u the up-and-down rate and $\varepsilon \approx 0.03$ for double chips and three-putts net of holed shots,

$$E_{\text{ARG}} \approx 2u + 3(1 - u) + \varepsilon = 3 - u + \varepsilon$$

Scratch inputs: 63% inside 25 yards, 32% from 25–50 yards, 37% sand saves. Chips from rough add 0.20.

A recovery shot (blocked by trees) is a punch-out that gains about 100 yards into a fairway/rough mix, with $d' = \max(d - 100, 15)$:

$$E_X(d) = 1.10 + 0.6\,E_F(d') + 0.4\,E_R(d')$$

5. Tee shots on par 4s and 5s

For a club with distance D and lateral standard deviation $\sigma = sD$, on a corridor of width w, a normal miss distribution gives the probability of staying in play

$$P_{\text{in}} = 2\,\Phi\!\left(\frac{w/2}{\sigma}\right) - 1$$

Misses split evenly left and right. A tree-side miss becomes a recovery lie. An out-of-bounds miss is stroke and distance, which means replaying from the tee at the same expectation E; solving the recursion,

$$E = 1 + P_{\text{in}}E_{\text{in}} + P_{\text{tree}}E_X + P_{\text{OB}}(1 + E) \quad \Longrightarrow \quad E = \frac{1 + P_{\text{in}}E_{\text{in}} + P_{\text{tree}}E_X + P_{\text{OB}}}{1 - P_{\text{OB}}}$$

with $E_{\text{in}} = 0.7\,E_F(L) + 0.3\,E_R(L)$ and $L = \max(\text{hole length} - D, 15)$. The hole's value is the minimum over the scratch club set, i.e. a scratch golfer picks the right club. Par 3s use E_F at the hole's yardage from the tee.

Club	Distance (yds)	σ (% of distance)	Basis
Driver	262	7.5	Shot Scope P-avg 285 $\times$ 0.92; Arccos big-miss rate implies $\sigma \approx$ 20–22 yds
3-wood	245	8.0	Shot Scope P-avg 261–267 $\times$ 0.92; fairway % barely better than driver
3-hybrid	203	7.5	Shot Scope scratch average 197, P-avg 219; dispersion assumed
5-iron	185	7.0	Shot Scope P-avg 200 $\times$ 0.92; dispersion assumed

Hole	Yards	Trouble	Scratch best club	Raw E
1	356	OB both sides	5-iron	4.55
3	262	OB right	3-wood	3.89
6	397	Trees	Driver	4.37
9	301	Trees	Driver	3.95
11	333	Trees	Driver	4.10
14	260	Trees	3-wood	3.72
16	321	OB right	Driver	4.30

6. Calibration to the course rating

The raw hole values sum to 64.15 at Fox Creek. The target is the course rating, 59.9, which is a scratch golfer's good day. The gap is spread over holes with weights that sum to one for any λ (171 is the sum of 1 to 18):

$$a_h = \left(\mathrm{CR} - \textstyle\sum_h E_h^{\mathrm{raw}}\right)\omega_h, \qquad \omega_h = \frac{1-\lambda}{18} + \lambda\,\frac{19-\mathrm{SI}_h}{171}$$

With $\lambda = 0$ every hole gets $a_h = -0.236$. A negative adjustment means the course plays easier than a generic course with the same hole lengths once tightness is already in the tee model; about 2.5 of the 4.25 gap is the good-day shift, the rest is short, wedge-length par 3s.

Within a hole, each position gets an adjusted expectation $E' = E + a_h\,w$ with

$$w = \begin{cases} 1 & \text{tee} \\ 0 & \text{on the green} \\ \mathrm{clip}\left[\dfrac{E - g_0}{E_{\mathrm{tee}} - g_0},\ 0,\ 1\right] & \text{otherwise} \end{cases}, \qquad g_0 = 1.85$$

Summing SG over a hole telescopes to $E_h^{\mathrm{raw}} + a_h -$ strokes regardless of w, so over eighteen holes total SG is exactly CR $-$ score. Because $w = 0$ on the green, putting strokes gained never carries a course adjustment: zero in the putting column means putting like the published scratch data.

7. Worked example: hole 1, driver to 102 yards in the fairway

Shot	E'(start)	E'(end)	SG
Drive, tee to fairway at 102 yds	$4.55 - 0.236 = 4.31$	$3.03 - 0.236 \times 0.44 = 2.93$	+0.38
Wedge, 102 yds to 22 ft	2.93	$E_G(22) = 1.95$	−0.02
First putt, 22 ft to 3 ft	1.95	1.06	−0.11
Second putt, holed	1.06	0	+0.06

The drive beats the scratch baseline, a 5-iron layup, by about 0.4 strokes. The weight on the 102-yard position is $w = (3.03 - 1.85)/(4.55 - 1.85) = 0.44$; the putts carry no adjustment.

8. What is data and what is assumption

Component	Source	Status
Putting make and three-putt rates by distance	Shot Scope (0 hcp), Arccos	Published
Approach proximity incl. misses by band; overall GIR	Shot Scope	Published
Split of each band into GIR and on-green proximity	Fitted to the published band averages	Derived

Component	Source	Status
Up-and-down inside 25 and 25–50 yds; sand save	Shot Scope	Published
Scratch club distances	Shot Scope P-avg × 0.92	Derived
Driver and 3-wood dispersion	Arccos big-miss rate; Shot Scope fairway %	Derived
Hybrid and long-iron dispersion; rough and bunker factors; recovery and layup rules; κ	—	Assumption

Scratch sits behind tour by roughly +0.02 to +0.13 strokes on putts, +0.05 to +0.15 around the green and +0.2 to +0.3 on approaches, consistent with Broadie's finding that approach play is the largest scratch-to-tour gap. The planned extension is credibility weighting: after n logged rounds, blend the model's hole values with observed hole averages using a Bühlmann-style weight $Z = n/(n + k)$, so assumed parameters give way to measured ones as data accumulates.

Code, tests and the spreadsheet: github.com/Eruven/golf-sg-model.

Sources

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